SENG 305 Operating Systems

Programming Assignment 2: Synchronization

Goal: To understand how programing languages support synchronization: the benefits and drawbacks of their design choices.

Programing language: C/C++

OS: Linux

- Implement the Producer-Consumer problem in C++ using (1) Pthreads and using (2) OpenMP.
  - The Producer-Consumer implementation should have a bounded buffer and multiple concurrent producers and consumers.
  - For PThreads: use mutex(es) and condition variables
  - Explain and justify the synchronization mechanisms used in each implementation.
- Compare the two approaches in terms of:
  - Ease of use
  - Clarity of code
  - Deadlock avoidance
- Notes: Make sure the implementations are a correct implementation of the Producer-Consumer problem.